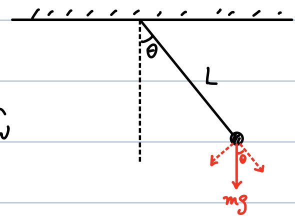
• Application of oscillatory motion

Simple Pendulum



$\odot\ \hat{w}$

$$\tau = I\alpha$$

$$\Rightarrow\ -mg\sin\theta\, L = mL^2 \frac{d^2\theta}{dt^2}$$

$$\Rightarrow\ \frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta$$

$$\theta \ll 1 \Rightarrow \frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$$

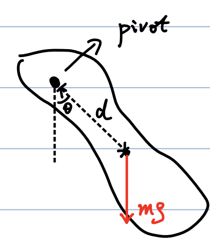
(Simple Harmonic oscillation)

$$\Rightarrow\ \boxed{\theta = \theta_{max}\cos(wt+\phi)}$$

$$w = \sqrt{\frac{g}{L}}$$

Period : $T = \frac{2\pi}{w} = 2\pi\sqrt{\frac{L}{g}}$

Physical Pendulum



$\odot\ \hat{w}$

$$\tau = I\alpha$$

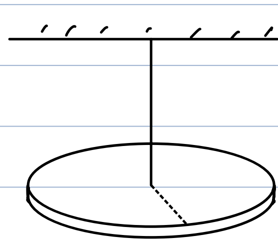
$$\Rightarrow\ -mgd\sin\theta = I\frac{d^2\theta}{dt^2}$$

$$\Rightarrow\ \frac{d^2\theta}{dt^2} = -\frac{mgd}{I}\sin\theta$$

$$\Rightarrow\ \theta = \theta_{max}\cos(wt+\phi)$$

$$w = \sqrt{\frac{mgd}{I}}$$

Torsion Pendulum (扭转摆)



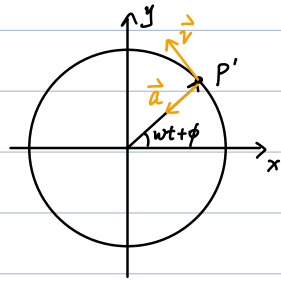
$$\tau = -k\theta$$

$$= I\ddot{\theta}$$

$$\Rightarrow\ \theta = \theta_{max}\cos(wt+\phi)$$

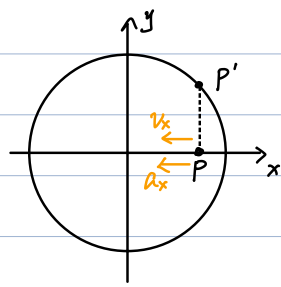
$$w = \sqrt{\frac{k}{I}}$$

Simple harmonic motion and uniform circular motion



P' : uniform circular motion

Project to x-axis :



$x = A\cos(\omega t+\phi)$

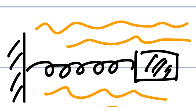
$\rightarrow$ simple harmonic motion

Project to y-axis : $y = A\sin(\omega t+\phi)$

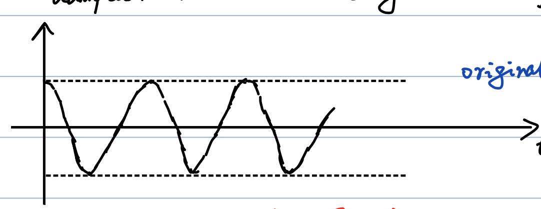
Notes :

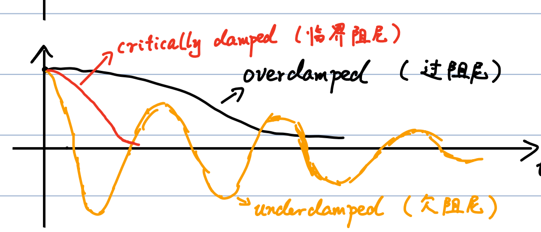
1. Projection of uniform circular motion $\Longleftrightarrow$ Simple harmonic motion
2. uniform circular motion $\Longleftrightarrow$ Combination of two simple harmonic motions

Damped oscillator



damped : motion reduced by external force.





e.g. fluid resistance.

$F_d = -bv$ (Low speed)

b : damping constant

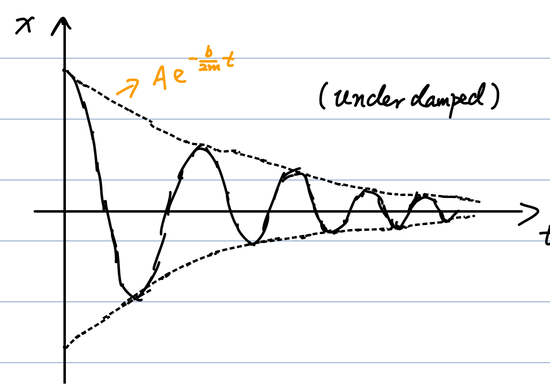
$F_{net} = -kx - bv = ma$

$\Rightarrow \quad m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$

Solution:

$$\boxed{x = Ae^{-\frac{b}{2m}t}\cos(\omega' t + \phi)}$$

$$\omega' \equiv \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2} \qquad \Big(\frac{b}{2m} < \underbrace{\sqrt{\frac{k}{m}}}_{\omega_0}\Big)$$



Overdamped:

$$x = A_1 \exp\left[\left(-\frac{b}{2m} + \sqrt{\left(\frac{b}{2m}\right)^2 - \omega_0^2}\right)t\right] \qquad \Big(\frac{b}{2m} > \sqrt{\frac{k}{m}}\Big)$$
$$+ A_2 \exp\left[\left(-\frac{b}{2m} - \sqrt{\left(\frac{b}{2m}\right)^2 - \omega_0^2}\right)t\right]$$

Critically damped:

$$\frac{b}{2m} = \sqrt{\frac{k}{m}}$$

$$\Rightarrow \omega' = 0$$

$$x = (A_1 + A_2 t)e^{-\frac{b}{2m}t}$$

Note:

critical damping $\longrightarrow$ fastest to restore to equilibrium position

$\Rightarrow$ good for devices such as ammeter.

Forced oscillation (driven oscillation)

$$F = -kx - bv + F_{ext}\cos(wt) = ma$$

$$\Rightarrow \quad \ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = \frac{F_{ext}}{m}\cos(wt)$$

Solution:

$$x = \underbrace{A' e^{-\frac{b}{2m}t}\cos(w't+\phi')}_{\substack{\text{transient solution} \\ (暂态解)}} + \underbrace{A\cos(wt-\phi)}_{\substack{\text{steady solution} \\ (稳态解)}}$$

where

$$\begin{cases} A = \dfrac{F_{ext}/m}{\sqrt{(w^2-w_0^2)^2 + (bw/m)^2}} \\ \tan\phi = \dfrac{bw/m}{w_0^2 - w^2} \end{cases}$$

Notes:

1. for $t \gg 1$, only steady solution survives
2. w: frequency of external force
3. ϕ: relative phase wrt F_{ext}
   (wrt: with respect to)

steady solution:

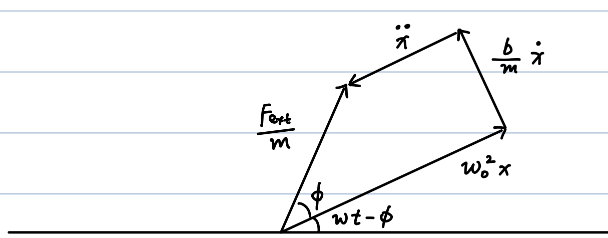
$$x = A\cos(wt-\phi)$$

$$\dot{x} = wA\cos(wt-\phi+\frac{\pi}{2})$$

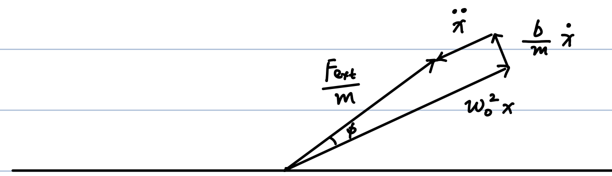
$$\ddot{x} = w^2 A\cos(wt-\phi+\pi)$$

$$\ddot{x} + \frac{b}{m}\dot{x} + w_0^2 x = \frac{F_{ext}}{m}\cos(wt)$$

as uniform circular motion:



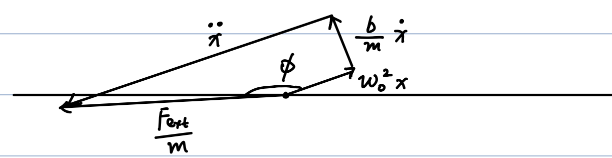
- slow drive ($\omega \ll \omega_0$)



$$\begin{cases} A \approx \dfrac{F_{ext}/m}{\omega_0^2} = \dfrac{F_{ext}}{k} \\ \phi \approx -\arctan \dfrac{b\omega/m}{\omega_0^2} \approx 0 \end{cases}$$

same phase as F_{ext}

- fast drive ($\omega \gg \omega_0$)

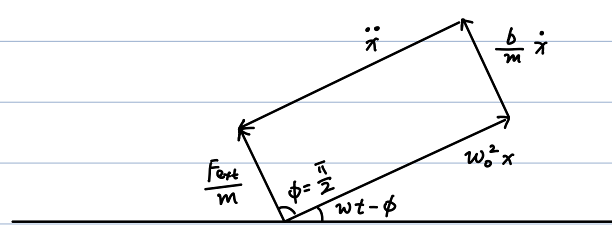


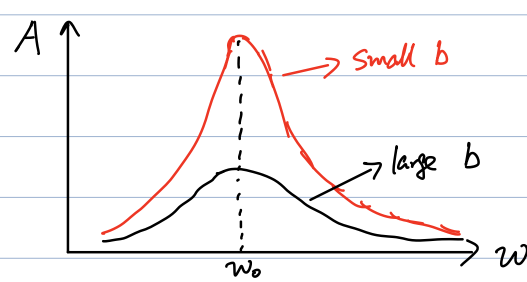
$$\begin{cases} A \approx \dfrac{F_{ext}/m}{\omega^2} \approx 0 \\ \tan\phi \approx \dfrac{b/m}{-\omega} \approx -0 \Rightarrow \phi \approx \pi \end{cases}$$

opposite phase as F_{ext}

- Resonance ($\omega = \omega_0$)

$$\tan\phi = \infty \Rightarrow \phi = \frac{\pi}{2}$$



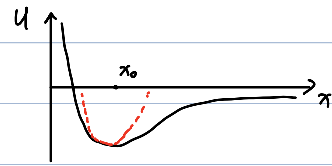


$A(\omega = \omega_0) \to \infty$ for $b \to 0$

Example. Lennard-Jones Potential

(Potential energy for a pair of neutral atoms or molecules)

$$U(x) = 4\epsilon\left[\left(\frac{\sigma}{x}\right)^{12} - \left(\frac{\sigma}{x}\right)^{6}\right]$$



Taylor expansion near x_0:

$$U(x) \approx U(x_0) + \frac{1}{2}\left.\frac{d^2U}{dx^2}\right|_{x_0}(x-x_0)^2$$

$$\left.\frac{dU}{dx}\right|_{x_0} = 0 \Rightarrow x_0 = 2^{1/6}\sigma \approx 1.122\sigma$$

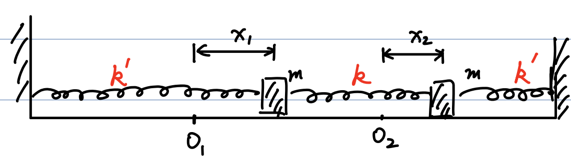
effective spring constant:

$$k \equiv \left.\frac{d^2U}{dx^2}\right|_{x_0} = \frac{72\epsilon}{x_0^2}$$

$$\Rightarrow \omega = \sqrt{\frac{k}{\mu}} \qquad \mu: \text{effective mass}$$

$$= \sqrt{\frac{72\epsilon}{\mu x_0^2}}$$

Two harmonic oscillators



$$\begin{cases} m\ddot{x}_1 = -k'x_1 - k(x_1 - x_2) \\ m\ddot{x}_2 = -k'x_2 + k(x_1 - x_2) \end{cases}$$

Generally a complicated problem.

Normal mode: $\begin{cases} \text{frequency "}\omega\text{" same for all oscillators} \\ \text{fixed relative phase} \end{cases}$

try $\begin{cases} x_1 = A_1\cos(\omega t + \phi_0) \\ x_2 = A_2\cos(\omega t + \phi_0) \end{cases}$

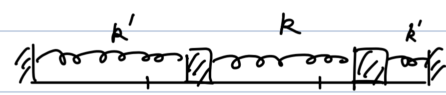
$$\Rightarrow \begin{cases} m\omega^2 A_1 = (k+k')A_1 - kA_2 \\ m\omega^2 A_2 = -kA_1 + (k+k')A_2 \end{cases}$$

$$\Rightarrow \begin{pmatrix} k+k' & -k \\ -k & k+k' \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = m w^2 \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$$

$m w^2$: eigenvalue of $\begin{pmatrix} k+k' & -k \\ -k & k+k' \end{pmatrix}$

Solution-1. translational mode

$$w = \sqrt{\frac{k'}{m}}, \qquad A_1 = A_2$$



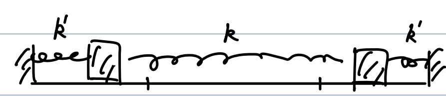
check: with $w = \sqrt{\frac{k'}{m}}$

$$\Rightarrow \begin{pmatrix} k & -k \\ -k & k \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = 0$$

$$\Rightarrow \quad A_1 = A_2$$

Solution-2. vibrational mode

$$w = \sqrt{\frac{2k+k'}{m}}, \qquad A_1 = -A_2$$



Method-2:

$$\begin{cases} m\frac{d^2}{dt^2}(x_1 + x_2) = -k'(x_1 + x_2) \\ m\frac{d^2}{dt^2}(x_1 - x_2) = -(2k + k')(x_1 - x_2) \end{cases}$$

$$\Rightarrow \begin{cases} x_1 + x_2 = C_1 \cos\left(\sqrt{\frac{k'}{m}}\, t + \phi_1\right) \\ x_1 - x_2 = C_2 \cos\left(\sqrt{\frac{2k+k'}{m}}\, t + \phi_2\right) \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = \frac{C_1}{2}\cos\left(\sqrt{\frac{k'}{m}}\, t + \phi_1\right) + \frac{C_2}{2}\cos\left(\sqrt{\frac{2k+k'}{m}}\, t + \phi_2\right) \\ x_2 = \frac{C_1}{2}\cos\left(\sqrt{\frac{k'}{m}}\, t + \phi_1\right) - \frac{C_2}{2}\cos\left(\sqrt{\frac{2k+k'}{m}}\, t + \phi_2\right) \end{cases}$$

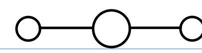
Notes: 1. $\{ C_1 = 2A_1,\ C_2 = 0 \} \longrightarrow$ translational mode

$\{ C_1 = 0,\ C_2 = 2A_2 \} \rightarrow$ vibrational mode

2. Overall motion: linear combination of normal modes

mode counting of N-atom molecule

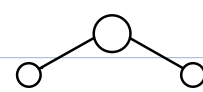
total modes: $3N$

- Linear molecule 

translation : 3 (x, y, z)

rotation : 2 (θ, φ)

Vibration : $3N - 5$

- Non linear molecule 

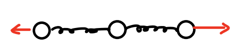
translation : 3

rotation : 3 (Euler angles, α, β, γ)

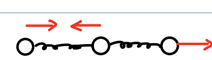
Vibration : $3N - 6$

Example. CO_2 ($N=3$, linear)

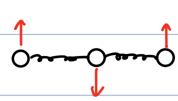
vibration : $3 \times 3 - 5 = 4$



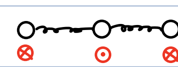
symmetric streching



asymmetric streching



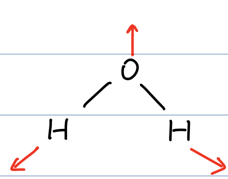
in-plane bendding



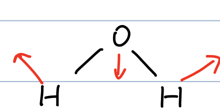
out-of-plane bending

Example. H_2O ($N=3$, nonlinear)

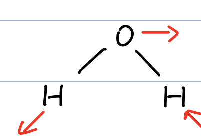
vibration : $3 \times 3 - 6 = 3$



symmetric streching



antisymmetric streching



bending